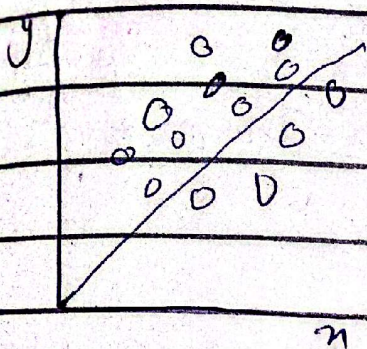
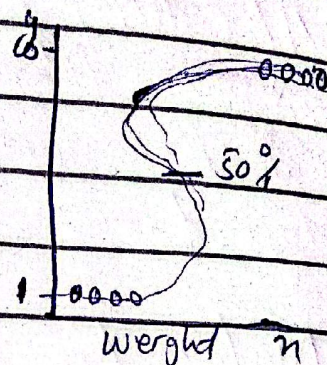


Logistic regression

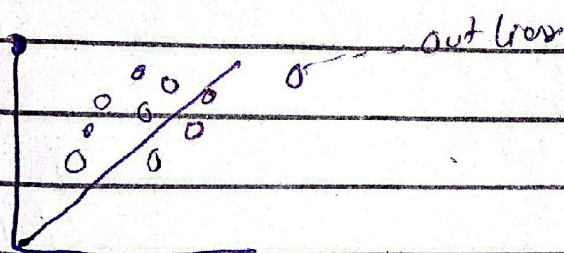
simple linear regl.



logistic regression



In logistic instead of making a straight line we use a "squash line"



Reasons: 1. Outliers influence the straight line.
2. Prediction can be less than 0 (< 0) or greater than 1 (> 1)

Note: To make the straight into S shaped curve we will use Sigmoid activation function.

$$\text{Form: } g = \frac{1}{1 + e^{-z}} \quad (0 + \theta(n))$$

$e \rightarrow$ eulers number $\rightarrow$ A mathematical constant 2.71818

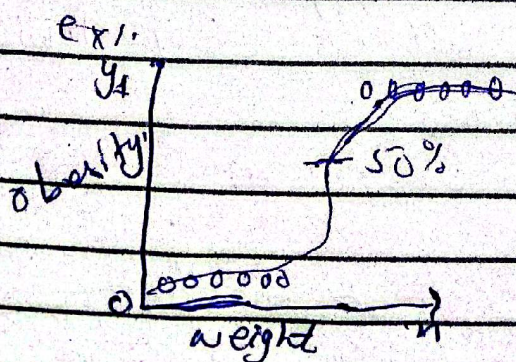
Note $\rightarrow$ If you find the best fit line, we need to use cost function.
cost function finds the residual error as we already discussed in Linear Reg.

After Applying cost function we will get one $\rightarrow$ error. then we see it on graph, and then we put it on gradient descent.

Note: In Logistic Regression we will use log loss function.
it also finds loss between data points but logistic regression work differently because we will ^{work on} probability.

Note: log loss function provide loss.
so same we will sum up all log loss.
then again we will use the gradient descent.

Use of:
 $\Rightarrow$ log loss function (Binary cross entropy)



1. Let's say predicted prob = 0.99
and actual is 1.
 $\Rightarrow$ very ~~low~~ less loss.

Let's say my predicted prob = 0.01 and
actual is 1.
 $\Rightarrow$ very high loss.

Note: log loss function. Apply penalty, means
if we get so much high loss then we apply
penalty $\rightarrow$ and penalty is calculate with
log loss function.
Formula:

$$\text{log loss} = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)}) \right]$$

m = number of samples

y = actual label (0-1)

$\hat{y}$ = predicted prob (0-1)